

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/323233272>

Using Plickers as an Assessment Tool in Health and Physical Education Settings

Article · February 2018

DOI: 10.1080/07303084.2017.1404510

CITATIONS

2

READS

1,756

2 authors:



[Lena Chng](#)

Ministry of Education, Singapore

4 PUBLICATIONS 2 CITATIONS

[SEE PROFILE](#)



[Rachel Gurvitch](#)

Georgia State University

37 PUBLICATIONS 350 CITATIONS

[SEE PROFILE](#)

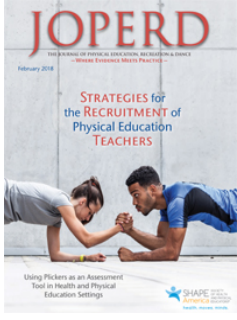
Some of the authors of this publication are also working on these related projects:



The effect of peer-assessment in opportunities to respond in a Badminton unit [View project](#)



Developmental Rubrics for Badminton [View project](#)



Using Pickers as an Assessment Tool in Health and Physical Education Settings

Lena Chng & Rachel Gurvitch

To cite this article: Lena Chng & Rachel Gurvitch (2018) Using Pickers as an Assessment Tool in Health and Physical Education Settings, Journal of Physical Education, Recreation & Dance, 89:2, 19-25, DOI: [10.1080/07303084.2017.1404510](https://doi.org/10.1080/07303084.2017.1404510)

To link to this article: <https://doi.org/10.1080/07303084.2017.1404510>



Published online: 16 Feb 2018.



Submit your article to this journal [↗](#)



Article views: 9



View related articles [↗](#)



View Crossmark data [↗](#)

USING *PLICKERS*



as an Assessment Tool

in Health and Physical Education Settings

LENA CHNG
RACHEL GURVITCH

Effective teachers are interested in measuring the learning outcomes of their lessons and units of instruction. They can only know if learning is taking place when they use some form of assessment. Assessment is the process of collecting data from students to use as the basis for evaluating the effectiveness of instruction, as well as for assigning grades (Imwold, Rider, & Johnson, 1982). Therefore, assessment becomes a critical component within the learning process because it provides opportunities to communicate expectations to students (Veal & Lund, 2013, p. 5).

When it comes to determining how well students have mastered a particular subskill or a body of knowledge, teachers have all

the traditional assessment tools at their disposal. They might employ multiple-choice or true/false questionnaires and short-answer items (Popham, 2008, p. 58). Written tests are the most common tools used to measure the cognitive and affective domains in physical education (Veal & Lund, 2013). Despite their popularity, there are numerous challenges in using written tests or written surveys in physical education settings — with the lack of time to type,

Lena Chng (lchng1@student.gsu.edu) is a doctoral student, and Rachel Gurvitch is an associate professor, in the Department of Kinesiology and Health at Georgia State University in Atlanta, GA.

print and administer written tests being one of the most common. In addition, teachers must spend time grading tests or surveys after their administration, which means their students do not get immediate feedback. Teachers are unable to use the information collected until after they have graded all tests and surveys. These could be reasons why physical education teachers limit their use of cognitive or affective assessments. Teachers usually check for understanding and how students feel as a class or group by asking questions at the beginning of the class or during closure. However, simply checking for understanding without systematic documentation of students' performance does not constitute an assessment (Veal & Lund, 2013).

If more states were to mandate the SHAPE America National Standards for K–12 Physical Education, regular and frequent assessments within the physical, cognitive and affective domains would become basic teacher expectations. The challenge is to help teachers find an easier, cost-effective, and time-effective way to assess their students' cognitive and affective performance during an instructional unit and toward the end of the unit. Therefore, the purpose of this article is to introduce the use of Plickers, a classroom response system that has the potential to make the assessment of the cognitive and affective domain in health and physical education easier (Krause, O'Neil, & Dauenhauer, 2017, p. 35).

A Classroom Response System

A classroom response system is a combination of hardware and software that facilitates teaching activities. A common example is the use of "clickers," where each student is given a handheld transmitter that beams a radio-frequency signal to a receiver attached to the teacher's computer. The students answer a multiple-choice question projected on the computer projector by selecting the answer of their choice on the clicker. The receiver on the teacher's computer collects the students' answers and produces a bar chart showing how many students chose each of the answer choices. The teacher then makes instructional choices based on the responses of the students (Bruff, 2017). Using this type of classroom response system is possible during physical and health education. However, it may not be ideal due to cost and the large number of students in these settings. Schools and organizations are not particularly keen on forking out big bucks for clunky devices used only for answering multiple-choice questions (Wan, 2012).

What Is Plickers?

Plickers is an online software that works like a classroom response system, and it is free. It is simple to use, and it allows teachers to collect real-time formative and summative assessment data without the need for any hardware devices. Plickers allows teachers to use response polling in class by giving each student a card (instead of a clicker), and then using a hand-held device (such as a phone or tablet) to scan the cards. Students' responses are then automatically collected and saved to the company's website (i.e., www.plickers.com). Teachers can use the data collected to check for understanding and to determine whether they should progress with the lesson or revisit certain concepts that were not mastered by most students.

Teachers using Plickers set up the test questions prior to instruction. These questions can be projected on a screen, handed out on a paper, or communicated verbally. Then students who are individually assigned a Plickers card at the start of the lesson will hold up

the Plickers card in a certain direction to indicate what they think is the correct answer. The teacher then scans their answers using the digital device. The Plickers program generates the response immediately. With the generated results, the teacher can address the question, check for understanding, or decide to move on with the lesson. The teacher can also download the results later to use them as a form of summative assessment. The use of the Plickers program gives all students the chance to participate and engage in learning without feeling self-conscious. Next is a step-by-step guide on how teachers can get started with the use of Plickers.

Getting Started with Plickers

The following equipment is needed to use Plickers in physical or health education settings: a laptop with an Internet connection, a screen projector, a screen or white wall, a handheld digital device (it can be an iPhone, Android phone, iPad, or a tablet) with an Internet connection, and a set of Plickers cards (one per student). Using the Plickers program for diagnostic, formative and summative assessment requires the teacher to follow eight steps.

Step 1: Download the App. The first step requires the teacher to download the Plickers application (app), which is free and available through the Apple or Android app store. Only the teacher needs to download the Plickers app on a handheld digital device (phone or a tablet).

Step 2: Create an Account. After the initial app download the teacher is asked to sign up and open a formal account on the Plickers homepage. The teacher is asked to provide their first and last name, email and a password to create an account on the website. Users can then manage their classes and create a library of questions on the website itself. This also allows teachers the flexibility to use the Plickers program on any device and with multiple classes, as all information is stored virtually.

Step 3: Print the Cards. The Plickers app can scan up to 63 responses at any one time. Depending on the number of students in the class, the teacher needs to print the number of cards that suits their instructional needs. Each card has a unique number that should be assigned to an individual student. Cards can be printed from www.plickers.com/cards. Each side on the card represents one of the letters A, B, C or D (see Figure 1). After the teacher has presented the specific question and its four possible answers represented by A, B, C or D, each student is asked to hold his or her own card so that the correct answer letter is at the top position. It is recommended that the cards be printed on the highest-quality paper or the heaviest card stock. This is to ensure the durability of the cards. If the teacher chooses to laminate the cards, he or she should use only matte laminates because the reflection of a glossy laminate may affect the ability of the cards to be properly scanned. These cards are also available for sale on Amazon.com. A deck of 40 cards with matte laminates costs \$20.

Step 4: Add the Class and Students. After the teacher has established an account, classes can be added on the "Classes" page by entering the basic information of the class. A color code may even be used for each class (see Figure 2). After the class is selected, students' names are added by just entering the names and hitting "Enter" (see Figure 3). Plickers automatically assigns the card numbers to students as their names are entered. The teacher can create multiple classes by using the "Add New Class" icon.

Step 5: Add Questions. Questions can be created in the Library of the webpage, or on the mobile app using the "+" icon. Questions can be in the form of multiple choice or true/false. Teachers

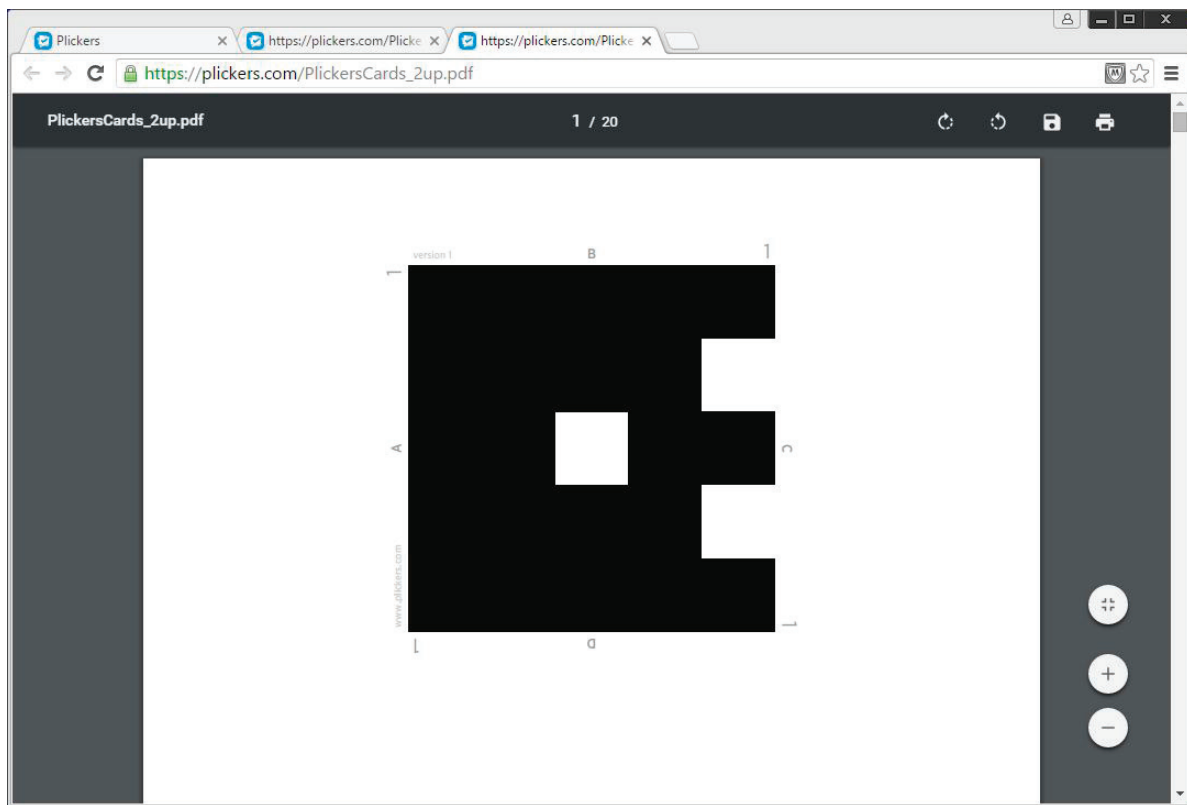


Figure 1.
Plickers card representing the letter B as the correct answer

© Plickers. Reproduced by permission of Plickers. Permission to reuse must be obtained from the original rightsholder.

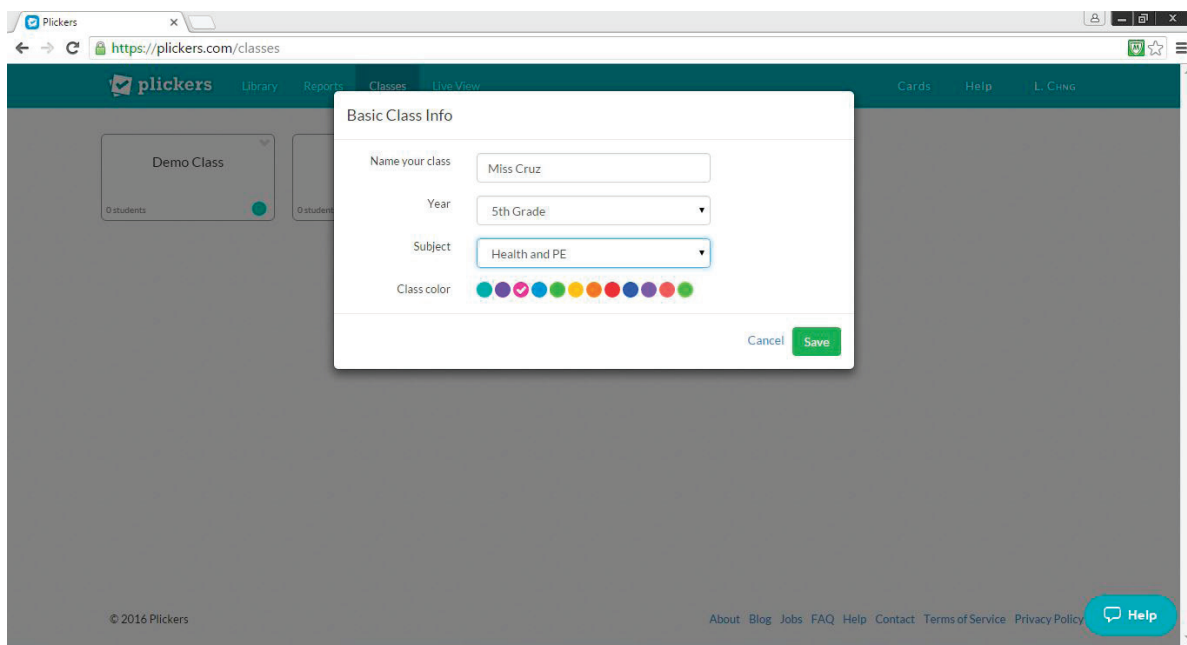


Figure 2.
Basic class information and class color

© Plickers. Reproduced by permission of Plickers. Permission to reuse must be obtained from the original rightsholder.

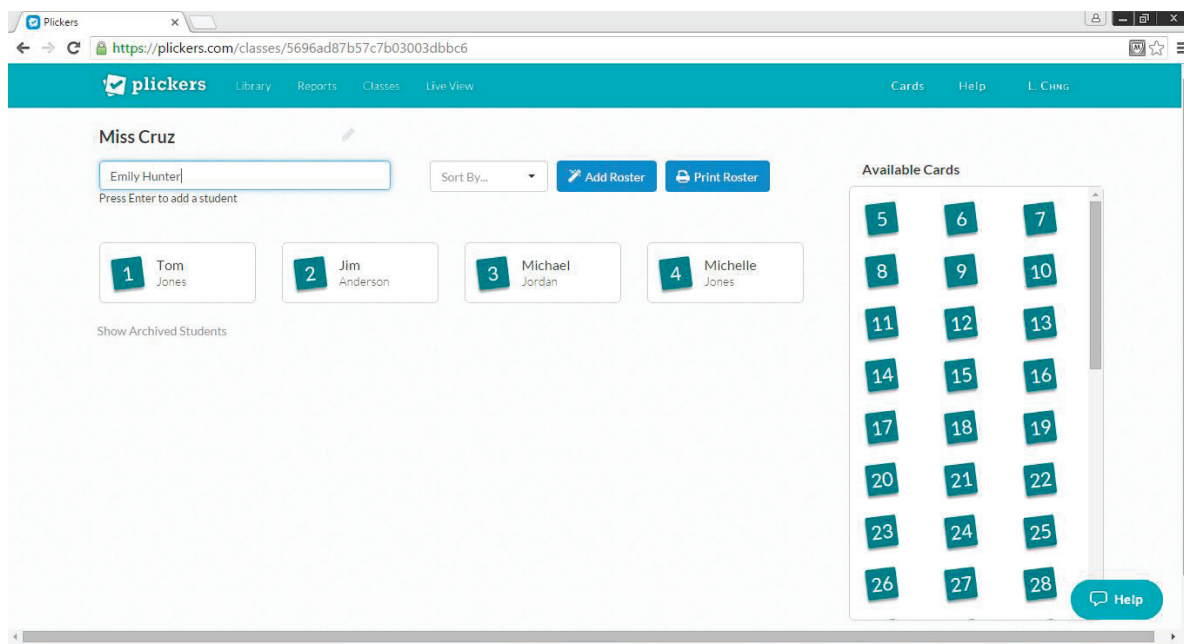


Figure 3.
Entering students' names

© Plickers. Reproduced by permission of Plickers. Permission to reuse must be obtained from the original rightsholder.

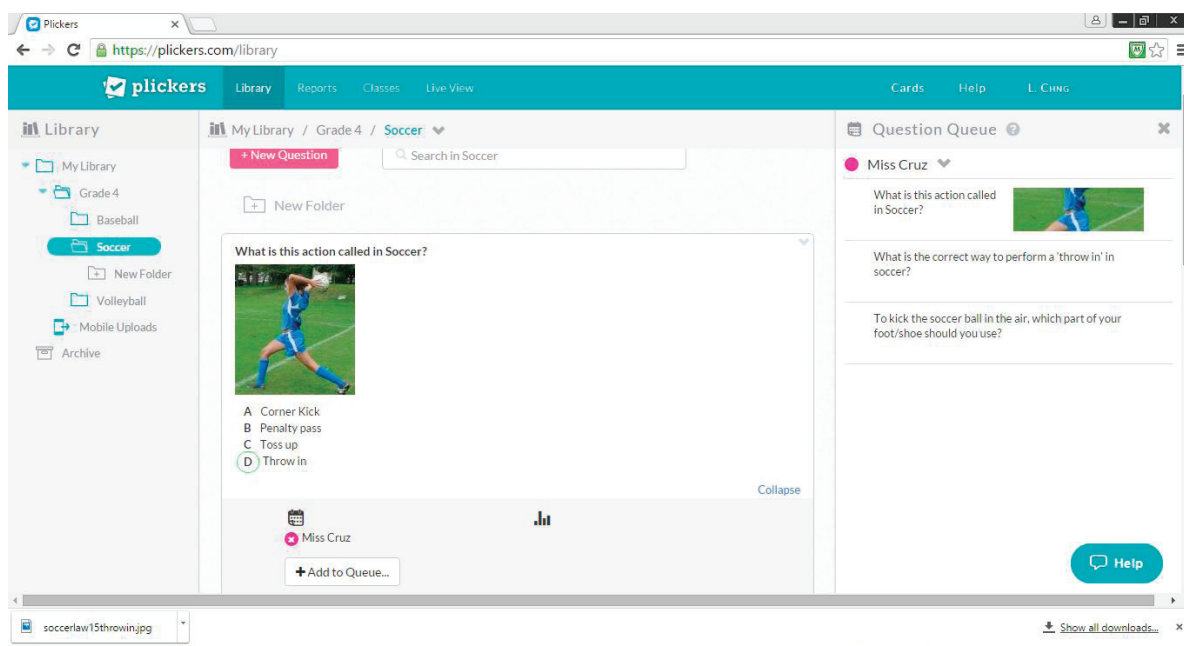


Figure 4.
How to organize questions

© Plickers. Reproduced by permission of Plickers. Permission to reuse must be obtained from the original rightsholder.

can also add images by clicking on the “Add Image” icon in the question box (see Figure 4). The folders in the Library can be arranged by level, instructional unit, or game type. New folders and subfolders can be created for easy access.

Step 6: Stay Organized. Before using it, the teacher must consider the management aspect of distributing and collecting cards. Since each card is unique to every student, it is important to make sure that each student holds his or her own card only. However,

the teacher can reuse the same cards for different classes. A teacher can use a card organizer for students to collect and return the cards in an organized manner (see Figure 5). For health education classes the teacher can print the Plickers cards on a piece of A4 size paper and ask students to paste their own Plickers card onto their class folder or health notebook. In that way, students have their card at all classes and use it every time the teacher needs to implement a quiz or survey. This minimizes the need to distribute the card every time the Plickers program is used. Regardless of the method used, it is recommended that each teacher come up with the plan that best works for a particular class.

Step 7: Implement Real-time Assessment. The teacher projects each question on the screen (see Figure 6). Students read the question and select their response by holding the card in a certain direction to indicate their response. There are small letter markings on the cards so that students know how to hold the card to indicate their response. At the same time the teacher taps the camera icon on the mobile app to scan the students' responses (see Figure 7) and instantaneously shows on the screen whose responses have been captured. Students whose responses have been captured put their cards down. The mobile app will also be able to indicate to the teacher the students whose responses are not captured, which is also helpful in determining attendance.

Scanning 60 students takes approximately 10 seconds. It is suggested that the teacher stand at the front of the class where he or she can see all the students within a small angle range to facilitate the speed of capturing the responses. A larger angle range may mean more time needed to scan. The teacher does not need to grade the answers; the application will provide a bar chart to tally students' responses and show the correct answer. Students receive immediate feedback, and their responses are recorded.



Figure 5.
Card organizer

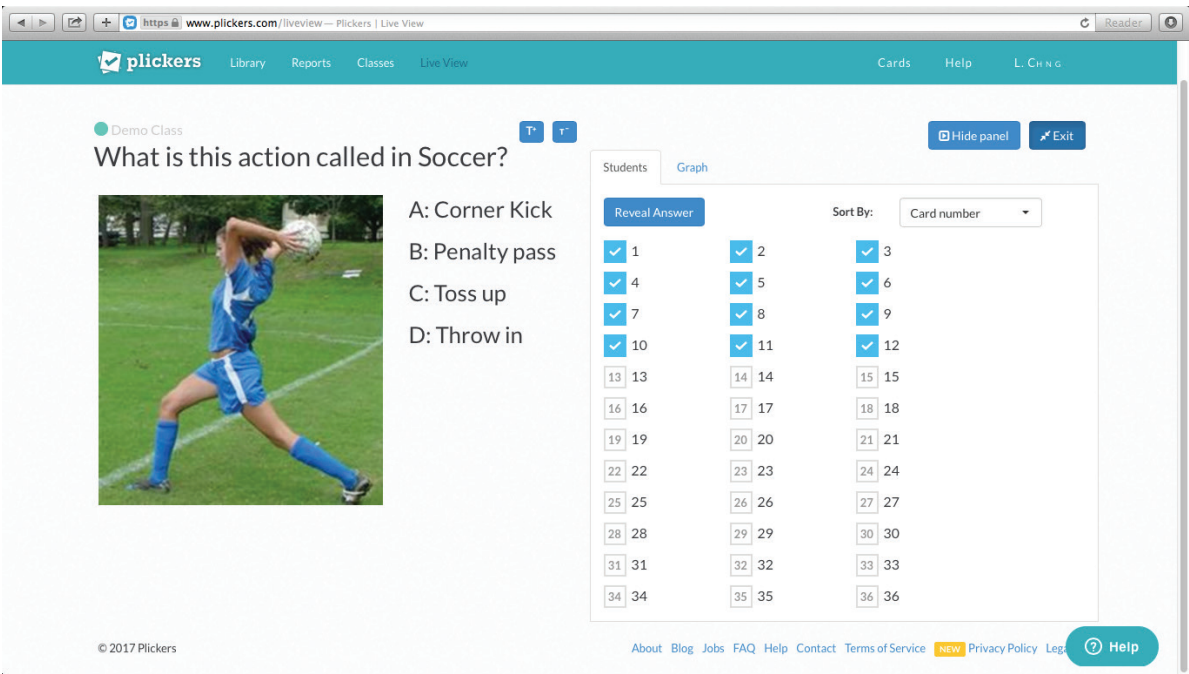


Figure 6.
Projection on wall

© Plickers. Reproduced by permission of Plickers. Permission to reuse must be obtained from the original rightsholder.



Figure 7.
Scanning of responses

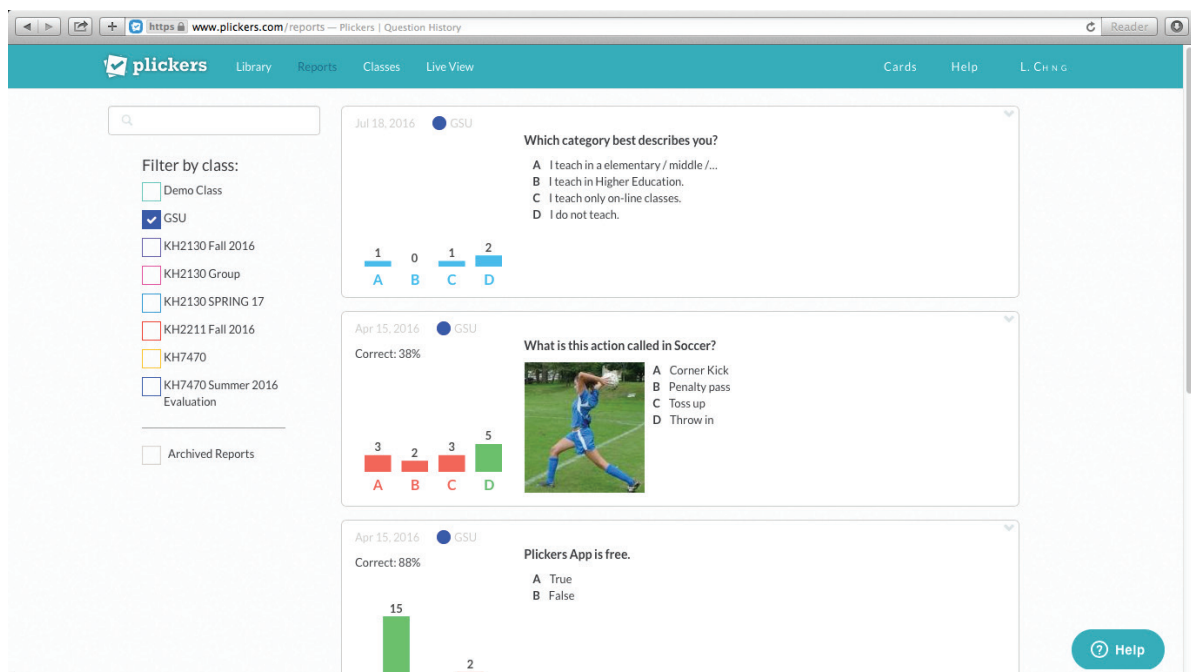


Figure 8.
Live feedback screen

© Plickers. Reproduced by permission of Plickers. Permission to reuse must be obtained from the original rightsholder.

Step 8: View Results. Teachers can also use the “Live View” tab on the website to display classroom results for students (see Figure 8). Students will be able to see immediately the number of students who got the answers right. The teacher can at the same time

go through the answer with the class, to correct any misconceptions. The use of the Plickers program allows the teacher to know immediately if the knowledge or concepts have been mastered and to plan the next course of action in their teaching. All results can

also be viewed on the Reports page to identify concepts to review and celebrate skills mastered.

Applications of Plickers

Like any classroom response system, Plickers can be used in many scenarios. For example, a teacher can start the lesson by administering a pop quiz on the previous concepts taught and, with the results collected, address any misconceptions and decide if the concept needs to be retaught or if the class can move on. In a physical education setting students do not usually carry with them any writing materials to the gym. Teachers can also use Plickers to administer a cognitive test on the rules of the game, critical elements of a skill, tactics of the game, and so on, and the results can be used as part of a summative grade. Teachers can also set survey questions on how students feel toward the lesson, their friend's attitude, and their own effort. This will help in saving time to collate the results. The use of Plickers, or any classroom response system, facilitates data collection. The value of the data is still dependent on the reliability and validity of the assessment items.

Challenges

Like every use of technology in educational settings, there are some potential challenges that should be mentioned. The first is the dependability on app connectivity. The use of Plickers depends on the connection of the app, and if the app is not operable, there may be a need to restart or reinstall it. This, however, is uncommon. Another challenge is the time it takes to set up the program. It takes time initially to input student details and assign them to a Plickers card number. Time is also needed to create a library of questions to use during class. This may take between one to three hours initially (depending on the number of classes and the number of students in each class), but this is a one-time effort. The third challenge is that Plickers cards need to be in good condition in order for the scanner to read them. Laminating the cards may help, but the reflection from the laminate can make it difficult for the app to recognize the cards. Thus, as mentioned earlier, only matte laminates are recommended. The use of Plickers, like many other innovative technologies, may also have a learning curve for teachers and their students. It takes time and practice to learn how to use a new program, but once the teacher has tried it, he or she will gain confidence using it.

Advantages

Despite the aforementioned challenges in using the Plickers program, there are many advantages that make it worth considering for health and physical education settings. First, health and physical education teachers can use this program to perform formative and summative assessments of cognitive and affective domain performances electronically without the need for individual students to have a personal digital device. Hence, there is no need for students to wait for their devices to start up and log in. The Plickers program allows the teacher to provide students with immediate feedback. There is no down time caused by students having to wait for their teacher to grade the test before they know how well they did. This is an advantage that teachers learn to appreciate, in that they do not need to grade the test or tally the survey. The software

tallies the results immediately, and the teacher can save time (and paper!) on grading. This is especially true for teachers who have many students in their physical education classes.

The Plickers software also automatically stores data for each class. This documentation is important for both summative and formative assessments. The report can be easily viewed, printed and analyzed. Since the Plickers cards are only readable by the scanning device, students next to each other will not be able to see what their friends' answers are. This not only minimizes the possibility of cheating in a test, but students are less self-conscious if they are not sure about the right answer. A final advantage is that the use of Plickers can also help a teacher to take attendance, especially in a big class setting (60 students). Most of all, the Plickers software is available for use at no charge!

Conclusion

Plickers has been used in different academic subjects such as mathematics, science and even music. The authors of this article recommend the use of this software to benefit health and physical education teachers, especially when assessing the cognitive and affective domains. This software helps teachers to document students' learning, which is an important aspect of assessment. Teachers can use the data collected for reviewing and revising their instructional process and can provide feedback to students — both of which are critical elements of teaching and learning. Teachers who have been using the software admit that the pros outweigh the cons, and that the time spent learning to use this software was definitely worthwhile. The time invested in learning and preparing for its implementation should benefit teachers in the long run.

Acknowledgment

The authors would like to thank Ms. Mandy Moreno and Ms. Jana Polk of Fulton County Schools for their contributions to this article.

Disclosure

The authors are not affiliated in any way with Plickers, and there is no conflict of interest.

References

- Bruff, D. (2017). *Class response systems ("Clickers")*. Center for Teaching, Vanderbilt University. Retrieved from <https://cft.vanderbilt.edu/guides-sub-pages/clickers/>
- Imwold, C., Rider, R., & Johnson, D. (1982). The use of evaluation in public school physical education programs. *Journal of Teaching in Physical Education*, 2, 13–18.
- Krause, J. N., O'Neil, K., & Dauenhauer, B. (2017). Plickers: A formative assessment tool for K–12 and PETE professionals. *Strategies*, 30(3), 30–36.
- Popham, J. (2008). *Transformative assessment*. Alexandria, VA: Association for Supervision and Curriculum Development.
- Veal, M., & Lund, J. (2013). *Assessment-driven instruction in physical education*. Champaign, IL: Human Kinetics.
- Wan, T. (Sept. 18, 2012). Check this out: A comparison of audience response systems. *EdSurge News*. Retrieved from <https://www.edsurge.com/news/check-this-out-a-comparison-of-audience-response-systems>